

HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

LAST NAME

EXAMPLE 2: Chan Ai Beng

FIRST NAME

LAST NAME

CHAN AI BENG

EXAMPLE 3: Jamal bin Abas

FIRST NAME

LAST NAME

The image shows two 6x6 letter grids used for a word search puzzle. The first grid contains the letters J, A, M, A, L, B, I, N in its top row, and the second grid contains A, B, A, S in its top row. Both grids are filled with various other letters, some of which are highlighted in red.

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / /

Postcode: _____

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

CLASS
(optional)

A	K
B	L
C	M
D	N
E	O
F	P
G	Q
H	R
I	S
J	T

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval **C**, as shown.

A **B** **C** **D**



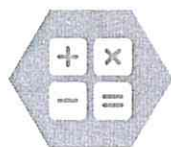
START

- 1 **A** **B** **C** **D**
- 2 **A** **B** **C** **D**
- 3 **A** **B** **C** **D**
- 4 **A** **B** **C** **D**
- 5 **A** **B** **C** **D**
- 6 **A** **B** **C** **D**
- 7 **A** **B** **C** **D**
- 8 **A** **B** **C** **D**
- 9 **A** **B** **C** **D**
- 10 **A** **B** **C** **D**

- 11 **A** **B** **C** **D**
- 12 **A** **B** **C** **D**
- 13 **A** **B** **C** **D**
- 14 **A** **B** **C** **D**
- 15 **A** **B** **C** **D**
- 16 **A** **B** **C** **D**
- 17 **A** **B** **C** **D**
- 18 **A** **B** **C** **D**
- 19 **A** **B** **C** **D**
- 20 **A** **B** **C** **D**

- 21 **A** **B** **C** **D**
- 22 **A** **B** **C** **D**
- 23 **A** **B** **C** **D**
- 24 **A** **B** **C** **D**
- 25 **A** **B** **C** **D**
- 26 **A** **B** **C** **D**
- 27 **A** **B** **C** **D**
- 28 **A** **B** **C** **D**
- 29 **A** **B** **C** **D**
- 30 **A** **B** **C** **D**

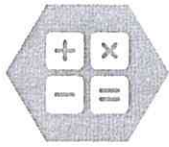
- 31 **A** **B** **C** **D**
- 32 **A** **B** **C** **D**
- 33 **A** **B** **C** **D**
- 34 **A** **B** **C** **D**
- 35 **A** **B** **C** **D**
- 36 **A** **B** **C** **D**
- 37 **A** **B** **C** **D**
- 38 **A** **B** **C** **D**
- 39 **A** **B** **C** **D**
- 40 **A** **B** **C** **D**



M | 2013 Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	B	A	C	D	B	A	C
2	B	B	B	A	B	D	C	C
3	D	C	D	A	B	C	C	B
4	D	D	A	C	D	B	B	C
5	C	C	D	C	C	A	D	D
6	C	C	B	D	A	A	B	A
7	B	A	A	D	D	D	A	D
8	C	A	A	A	D	B	A	A
9	A	C	B	C	C	C	B	A
10	B	A	D	B	B	D	D	B
11	B	B	C	D	D	A	A	C
12	C	D	C	A	A	C	D	D
13	D	D	B	D	D	A	C	A
14	A	A	D	A	A	A	A	C
15	B	C	B	C	A	C	D	B
16	B	B	B	A	A	D	B	C
17	D	D	A	A	B	A	D	D
18	D	C	C	B	A	D	D	B
19	B	C	C	A	A	D	C	D
20	A	B	B	A	C	B	B	A
21	D	C	D	A	A	C	B	B
22	B	D	C	B	C	D	C	D
23	A	D	A	D	D	B	B	C
24	C	A	C	B	D	C	C	D
25	A	D	B	A	A	B	C	B
26	B	C	B	B	D	B	C	A
27	C	B	D	C	A	A	D	A
28	C	C	D	D	C	C	D	B
29	B	C	A	B	B	A	B	C
30	D	A	B	A	B	B	C	C

(Please turn over)



M | 2013 Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	A	B	B	D	A	C	C	A
32	D	B	A	B	D	C	B	B
33	C	C	C	C	C	B	B	D
34	C	A	D	D	C	B	D	A
35	A	B	C	C	A	C	C	C
36	A	C	D	12 012	950	6 006 06	6 006 06	10 010
37	C	D	C	128	301	98 098	4 004 04	4 004 04
38	A	A	B	90 090	135	10 010	216	351
39	D	A	B	126	12 012	90 090	127	104
40	A	B	C	350	600	16 016	18 018	65 065

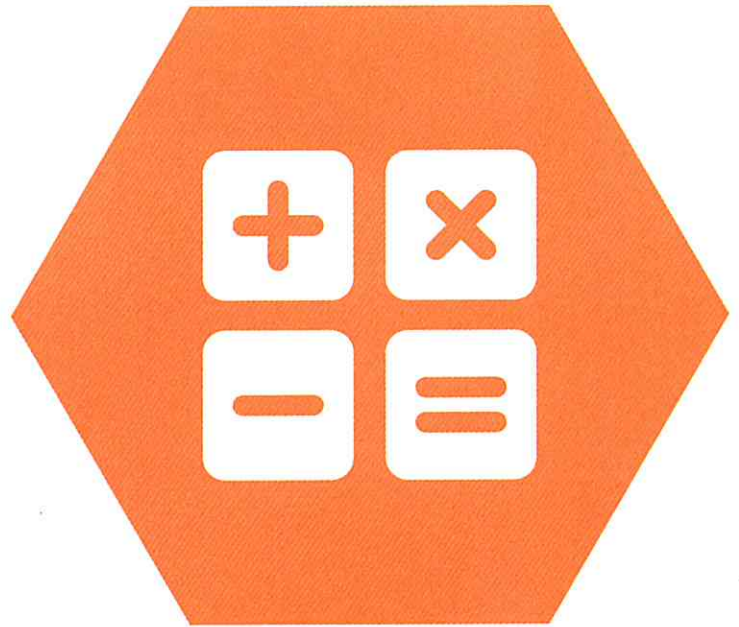
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UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER B



2013 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. Anna collected 8 water bottles.
Raj collected twice as many.



How many water bottles did Raj collect?

- (A) 24
- (B) 16
- (C) 8
- (D) 4

2. $\boxed{?} \times 4 = 28$

- (A) 6
- (B) 7
- (C) 8
- (D) 9

3. Danny tosses a coin that has two sides, A and B.

What is the chance that the coin lands on side A?

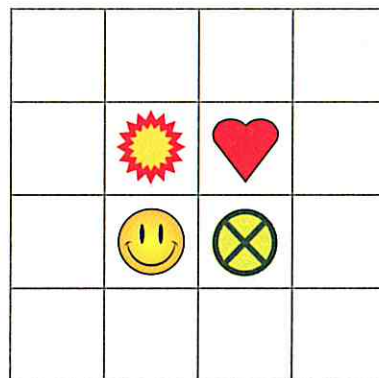
- (A) impossible
- (B) unlikely
- (C) fifty-fifty
- (D) certain

4. Sue read a series of 12 books. Each book took her 3 days to read.

How many days did it take Sue to read all the books?

- (A) 4
- (B) 12
- (C) 15
- (D) 36

5. Mia has some pictures.



Which picture is in the third row from the top and second column from the left?



(A)



(B)

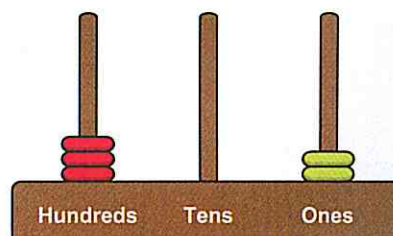


(C)



(D)

6. Danny used an abacus to show a number.



What number did Danny show?

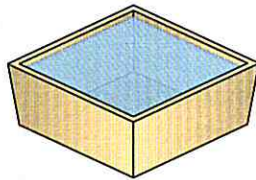
- (A) 32
- (B) 203
- (C) 302
- (D) 320

7. These identical boxes are filled to the top. Each box is filled with a different material.

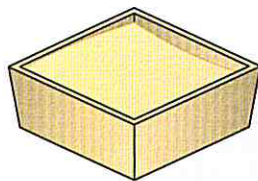
Which full box has the smallest mass?



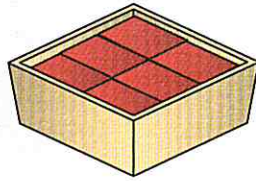
feathers
(A)



water
(B)



sand
(C)



bricks
(D)

8. A sack of rice weighs 56 kg.

How many 2 kg bags of rice can be filled from this sack?

- (A) 28
- (B) 56
- (C) 58
- (D) 112

9. Tony left home at 8:30 am to go on a day trip. He arrived back home at 3:40 pm.

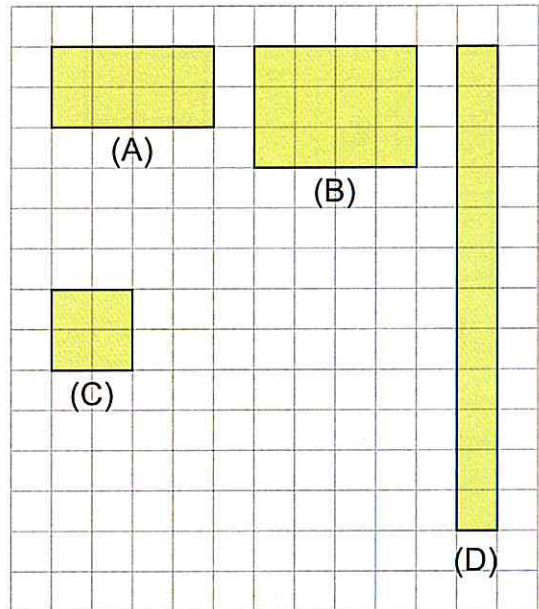


How long was Tony away from home?

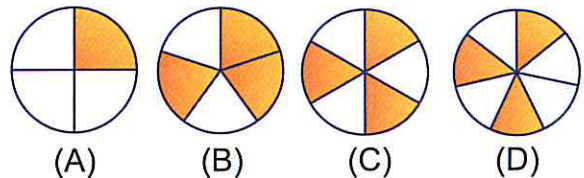
- (A) 4 hours 50 minutes
- (B) 5 hours 10 minutes
- (C) 7 hours 10 minutes
- (D) 11 hours 70 minutes

10. Anna drew some shapes on 1 cm grid paper.

Which shape has a perimeter of 12 cm?



11. Which circle has the largest area shaded?



12. Sue poured one litre of water into each of the containers shown.

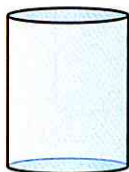
Which container has a capacity of less than one litre?



(A)



(B)

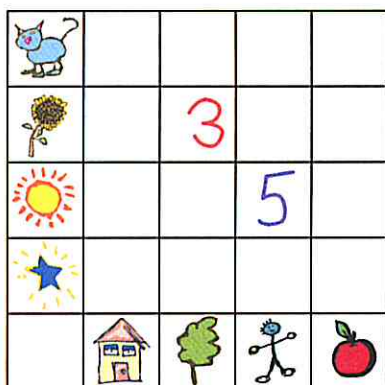


(C)



(D)

13. Noor is using this grid to represent the position of numbers.



In this grid the position of the number 3 is given as (flower, tree).

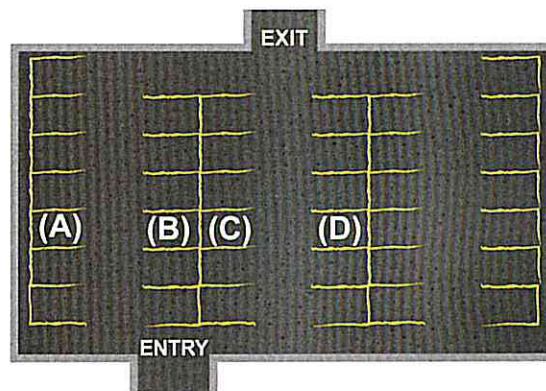
Using Noor's representation, what is the position of the number 5?

- (A) (sun, tree)
- (B) (tree, sun)
- (C) (boy, sun)
- (D) (sun, boy)

14. This is a map of a car park.

Raj drove his car into the ENTRY of the car park. He turned left and then right. He parked his car in the third parking space on his left.

In which parking space did Raj park his car?



15. Class A and Class B were going on a picnic. The teacher wanted to buy sandwiches for them.

The teacher collected the following information from the students.

Sandwich	Class A	Class B
Meat	10	15
Vegetarian	22	11
Total	32	26

How many more vegetarian sandwiches than meat sandwiches does the teacher need to buy?

- (A) 5
- (B) 6
- (C) 8
- (D) 16

16. Mia bought a one-kilogram bag of apples.
There were seven apples of similar size.



About how much does one apple weigh?

- (A) 1400 grams
- (B) 140 grams
- (C) 14 grams
- (D) 4 grams

17. Mia drew an arrow pointing north.



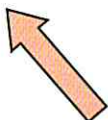
Which of these arrows is pointing north-east?



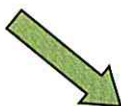
(A)



(B)



(C)

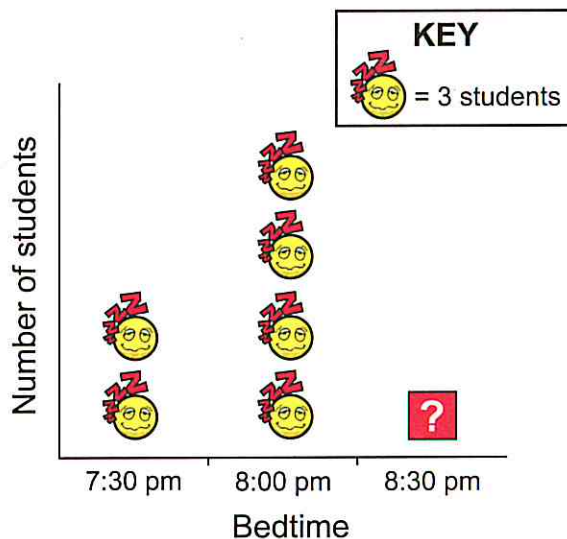


(D)

18. There are 21 students in Year 4. They all have a bedtime of 7:30, 8:00 or 8:30 pm.

The students started making a picture graph to show their bedtimes.

Year 4 Bedtime



The graph does not show the number of students with a bedtime of 8:30 pm.

How many students in Year 4 have a bedtime of 8:30 pm?

- (A) 0
- (B) 1
- (C) 3
- (D) 15

19. At an athletics carnival, every student entered at least one event.

The table shows the number of students in each event.

Event	Boys	Girls
Sprint	5	10
Long jump	10	5
High jump	2	3
Discus	3	2

What is the smallest number of students who were at the athletics carnival?

- (A) 5
(B) 15
(C) 20
(D) 40

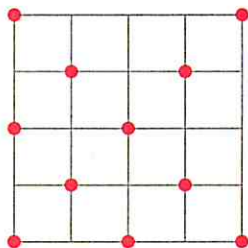
20. Jess wrote this number sentence

$$2 \times \text{😊} - \text{😊} = 2 \times \text{★} + \text{😊}$$

Which option makes this number sentence true?

- (A) 😊 = 1, ★ = 1
(B) 😊 = 2, ★ = 0
(C) 😊 = 4, ★ = 3
(D) 😊 = 6, ★ = 2

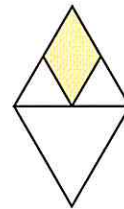
21. This square grid has 11 dots marked on it.



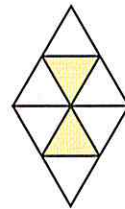
How many squares that have dots on ALL their corners can be drawn on this grid?

- (A) 3
(B) 4
(C) 5
(D) 6

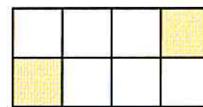
22. Which shape does NOT show one-quarter shaded?



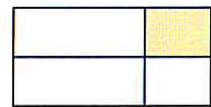
(A)



(B)

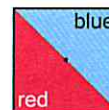


(C)

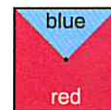


(D)

23. Four children each made a square out of cardboard. All the squares had the same area. Each child used two pieces of cardboard, one red and one blue.



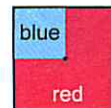
Anna



Mia



Raj

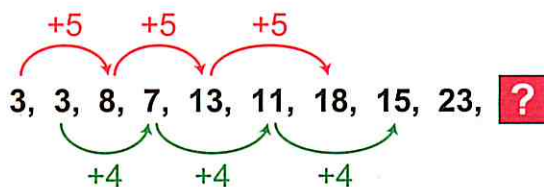


Danny

Which statement is true?

- (A) Raj's red piece has the same area as Danny's red piece.
(B) Mia's blue piece has the same area as Anna's blue piece.
(C) Anna's red piece has the same area as Danny's red piece.
(D) Mia's blue piece has the same area as Danny's blue piece.

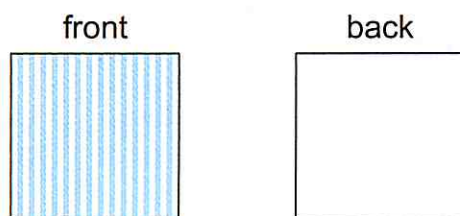
24. The diagram shows how Mia made a number pattern.



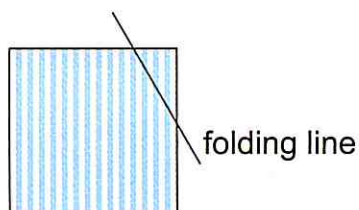
What number must **?** be?

- (A) 19
(B) 20
(C) 28
(D) 39

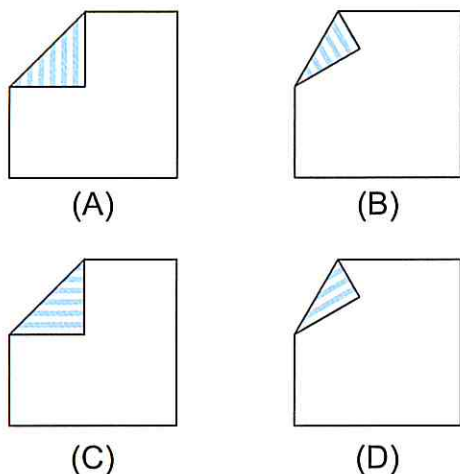
25. Sue has a piece of paper that has a pattern on the front, but is white on the back.



Sue folds a corner of the piece of paper towards the back.

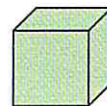


What would Sue's folded piece of paper look like from the back?



26. Danny has some cardboard squares that all have sides that are one metre long.

He made this closed box using six of his squares taped together. It is a cube that has a volume of one cubic metre.



Danny wants to make another closed box that has a volume of two cubic metres.

What is the smallest number of cardboard squares that he needs to make the other box?

- (A) 6
(B) 8
(C) 10
(D) 12

27. Sue is making a cheesecake.

Here is part of her recipe.



Sue does not have a measuring cup. She only has a teaspoon.

She uses the following conversion.

1 tablespoon = 3 teaspoons
1 cup = 16 tablespoons

How many teaspoons of sugar does Sue need for her recipe?

- (A) 16 (B) 12
(C) 4 (D) $\frac{1}{4}$

28. Anish traded all his old video games for new ones.



The game store gave Anish one new video game for every two old ones.

Then he was given four more new video games for his birthday.

Now Anish has ten new video games.

How many old video games did Anish have?

- (A) 3
- (B) 7
- (C) 12
- (D) 16

29. Edward has four cats.



Cato



Fluffy



Ginger



Snowball

Cato is older than Snowball. Fluffy is younger than both Cato and Ginger, but older than Snowball. Cato is younger than Ginger.

Which cat is the oldest?

- (A) Cato
- (B) Fluffy
- (C) Ginger
- (D) Snowball

30. Jess arranged some shapes into a repeating pattern.

Here are the first 12 shapes in the pattern.

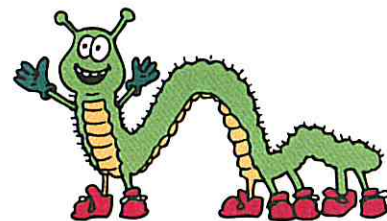


What will be the 46th shape in this pattern?

- (A)
- (B)
- (C)
- (D)

31. A caterpillar crawls 1 metre on the first day, 2 metres on the second day, and then 3 metres on the third day.

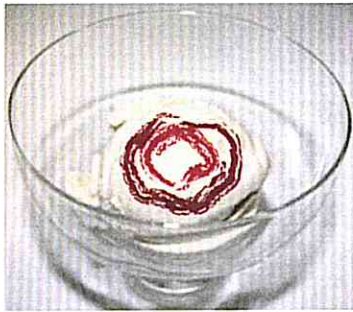
The caterpillar keeps crawling in this way until it crawls 10 metres on the tenth day.



How many metres in total will the caterpillar crawl in ten days?

- (A) 66
- (B) 55
- (C) 45
- (D) 10

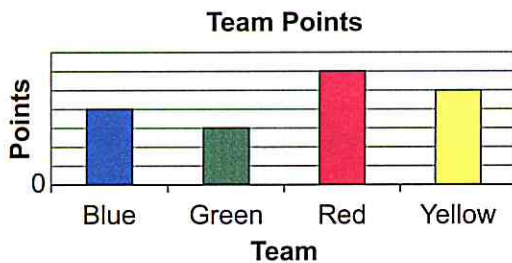
32. Jess makes a scoop of ice-cream that has a volume of 85 mL.



What is the largest number of scoops of this size that Jess could make from a 2 L tub of ice-cream?

- (A) 2
- (B) 23
- (C) 24
- (D) 42

33. This graph shows the points scored by four teams.

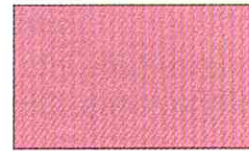


The Yellow team has 10 more points than the Green team.

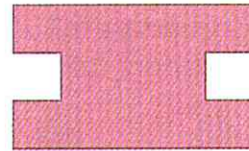
How many points does the Red team have?

- (A) 6
- (B) 15
- (C) 30
- (D) 60

34. Sue has a rectangle made of cardboard.



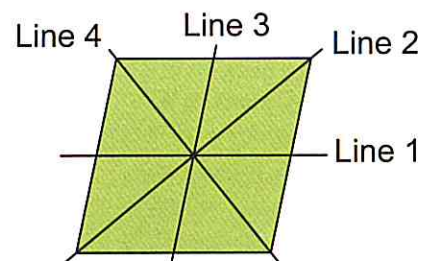
Sue cuts out a square from two sides as shown. Each square has a side one centimetre long.



Compared to the first shape, the perimeter of the new shape is

- (A) longer by four centimetres.
- (B) longer by six centimetres.
- (C) shorter by two centimetres.
- (D) the same length.

35. Anna drew a shape and four lines as shown.



Which of these lines are lines of symmetry for the shape?

- (A) Line 1 and Line 3 only
- (B) Line 2 and Line 4 only
- (C) all four lines
- (D) none of the four lines

36. Danny uses a cup and a jug to mix juice. It takes 2 cups of juice to fill the jug.

He pours orange juice into the jug until it is half full. Then he adds half a cup of apple juice.

How full is the jug now?

- (A) one-quarter full
- (B) two-quarters full
- (C) three-quarters full
- (D) four-quarters full

37. Anna found a page from a 2008 calendar.

May 2008						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
		13	14	15	16	17

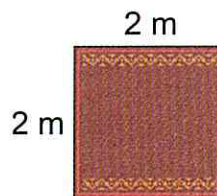
This note was on the page.



When did Adriana go to France?

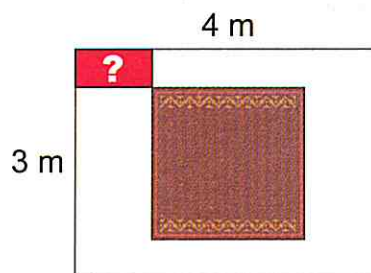
- (A) Thu 22 May 2008
- (B) Thu 29 May 2008
- (C) Wed 4 June 2008
- (D) Thu 5 June 2008

38. Mia bought a 2 m by 2 m rug.



Mia placed the rug exactly in the middle of the room.

The room measures 3 m by 4 m.

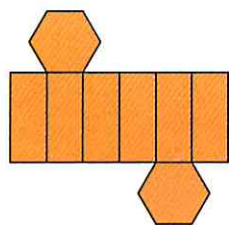


Mia placed a smaller rug that fits in the left top corner.

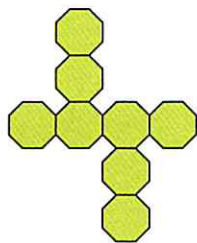
What is the area of the smaller rug?

- (A) $\frac{1}{2}$ square metre
- (B) 1 square metre
- (C) $1\frac{1}{2}$ square metres
- (D) 2 square metres

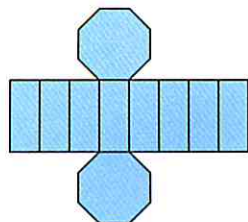
39. Which of the following is a net of a 3D shape with 8 faces?



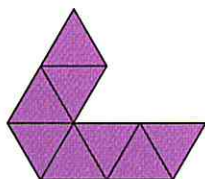
(A)



(B)



(C)



(D)

40. Sue has four flowers. She can arrange her flowers equally between vases in three different ways.

(1) one vase – four flowers



(2) two vases – two flowers in each



(3) four vases – one flower in each



Raj, Danny, Haji and Edward have some flowers.

Who has the largest number of ways to arrange his flowers equally between vases?

Edward
17 flowers



(A)

Haji
16 flowers



(B)

Danny
15 flowers



(C)

Raj
14 flowers



(D)

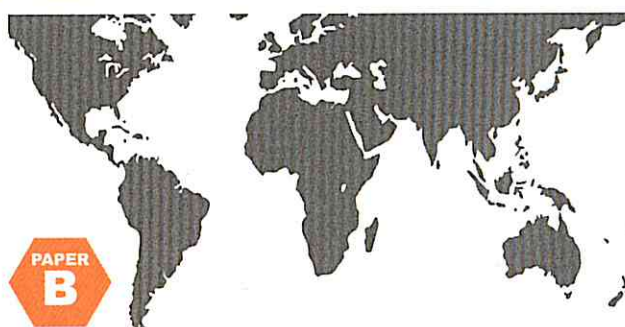
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The following year levels should sit THIS Paper:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 5
Singapore	Primary 3
South Africa	Grade 4



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